

MATH0005 Algebra 1 — Practice Problems & Solutions

Problem 1 — Linear Systems with a Parameter

Let A be the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & \lambda \\ 1 & 4 & \lambda^2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

(a) For which values of $\lambda \in \mathbb{R}$ does the system $A\mathbf{x} = \mathbf{b}$ have: no solution, exactly one solution, infinitely many solutions?

(b) For the value(s) of λ giving infinitely many solutions, find the general solution and express it as a particular solution plus the fundamental solutions of $A\mathbf{x} = \mathbf{0}$.

Solution

Row reduce the augmented matrix $[A|\mathbf{b}]$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 5 & \lambda & 3 \\ 1 & 4 & \lambda^2 & 2 \end{array} \right)$$

$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & \lambda - 2 & 1 \\ 0 & 2 & \lambda^2 - 1 & 1 \end{array} \right)$$

$R_3 \rightarrow R_3 - 2R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & \lambda - 2 & 1 \\ 0 & 0 & \lambda^2 - 2\lambda + 3 & -1 \end{array} \right)$$

Now $\lambda^2 - 2\lambda + 3 = (\lambda - 1)^2 + 2 > 0$ for all real λ . The $(3,3)$ entry is **never zero**.

Part (a) Since the bottom-right pivot $\lambda^2 - 2\lambda + 3 \neq 0$ for all $\lambda \in \mathbb{R}$, the matrix always has 3 pivot columns and the system is always consistent with a unique solution.

- **No solution:** never.
- **Exactly one solution:** for all $\lambda \in \mathbb{R}$.
- **Infinitely many solutions:** never.

Part (b) Since there are no values of λ giving infinitely many solutions, part (b) is vacuous. For any λ , $N(A) = \{\mathbf{0}\}$ and there are no fundamental solutions.

Problem 2 — Subspaces, Rank-Nullity, and Linear Maps

Let $T : \mathbb{R}^5 \rightarrow \mathbb{R}^4$ be a linear map and let $U = \ker T$.

(a) Suppose $\dim(\operatorname{Im} T) = 3$. Find $\dim U$ using rank-nullity. Show that U cannot be spanned by fewer than 2 vectors.

(b) Now suppose in addition that $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^5$ satisfy $T\mathbf{v}_1 = T\mathbf{v}_2 \neq \mathbf{0}$. Show that $\mathbf{v}_1 - \mathbf{v}_2 \in U$ and that $\{\mathbf{v}_1 - \mathbf{v}_2\}$ can be extended to a basis of U .

(c) Prove or disprove: there exists a linear map $S : \mathbb{R}^4 \rightarrow \mathbb{R}^5$ such that $T \circ S = \operatorname{Id}_{\mathbb{R}^4}$.

Solution

Part (a) By the rank-nullity theorem applied to $T : \mathbb{R}^5 \rightarrow \mathbb{R}^4$:

$$\dim(\ker T) + \dim(\operatorname{Im} T) = 5 \implies \dim U = 5 - 3 = 2.$$

Suppose for contradiction that $U = \operatorname{span}\{\mathbf{f}\}$ for some single vector $\mathbf{f}$. Then $\{\mathbf{f}\}$ is a spanning sequence of length 1 for U . But $\dim U = 2$, so by the Steinitz exchange lemma any spanning sequence of U has length at least 2 — contradiction. Hence U cannot be spanned by fewer than 2 vectors. $\square$

Part (b) $\mathbf{v}_1 - \mathbf{v}_2 \in U$: Since T is linear,

$$T(\mathbf{v}_1 - \mathbf{v}_2) = T\mathbf{v}_1 - T\mathbf{v}_2 = \mathbf{0}.$$

Hence $\mathbf{v}_1 - \mathbf{v}_2 \in \ker T = U$.

Extension to a basis: Since $T\mathbf{v}_1 \neq \mathbf{0}$, we have $\mathbf{v}_1 \notin U$, so $\mathbf{v}_1 - \mathbf{v}_2 \neq \mathbf{0}$ (if it were zero then $\mathbf{v}_1 = \mathbf{v}_2$, but then $T\mathbf{v}_1 = T\mathbf{v}_2$ would imply $\mathbf{v}_1 \in U$, contradicting $T\mathbf{v}_1 \neq \mathbf{0}$). So $\{\mathbf{v}_1 - \mathbf{v}_2\}$ is a non-zero linearly independent set in U . Since $\dim U = 2$, by the extension lemma it can be extended to a basis of U by adding one more vector. $\square$

Part (c) Claim: No such S exists.

Suppose for contradiction that $S : \mathbb{R}^4 \rightarrow \mathbb{R}^5$ satisfies $T \circ S = \text{Id}_{\mathbb{R}^4}$. Then for any $\mathbf{y} \in \mathbb{R}^4$, $T(S(\mathbf{y})) = \mathbf{y}$, so T is **surjective**, meaning $\dim(\text{Im } T) = 4$.

But we are given $\dim(\text{Im } T) = 3 \neq 4$. Contradiction. So no such S exists. $\square$

Note: A right inverse ($T \circ S = I$) requires T to be surjective. A left inverse ($S \circ T = I$) requires T to be injective. These are distinct conditions — a function can be non-injective and still possess a right inverse.

Problem 3 — Gram–Schmidt and Orthogonal Projections

Let $W = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ where

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \\ 0 \end{pmatrix}.$$

(a) Apply Gram–Schmidt to $\{\mathbf{u}_1, \mathbf{u}_2\}$ to obtain an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for W .

(b) Find the coordinate vector of $\mathbf{b} = (1, 0, 0, 0)^T$ with respect to the orthonormal basis, and write $\mathbf{b}$ as a sum of a vector in W and a vector orthogonal to W .

(c) Hence find the vector in W closest to $\mathbf{b}$, and compute the distance from $\mathbf{b}$ to W .

Solution

Part (a): Gram–Schmidt Set $\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 0, 1)^T$, so $\|\mathbf{v}_1\|^2 = 3$.

$$\mathbf{u}_2 \cdot \mathbf{v}_1 = 2 + 1 + 0 + 0 = 3, \quad \text{proj}_{\mathbf{v}_1}(\mathbf{u}_2) = \frac{3}{3}\mathbf{v}_1 = (1, 1, 0, 1)^T.$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \text{proj}_{\mathbf{v}_1}(\mathbf{u}_2) = (2, 1, 1, 0)^T - (1, 1, 0, 1)^T = (1, 0, 1, -1)^T.$$

Check: $\mathbf{v}_1 \cdot \mathbf{v}_2 = 1 + 0 + 0 - 1 = 0$. Also $\|\mathbf{v}_2\|^2 = 1 + 0 + 1 + 1 = 3$.

Normalising:

$$\mathbf{e}_1 = \frac{1}{\sqrt{3}}(1, 1, 0, 1)^T, \quad \mathbf{e}_2 = \frac{1}{\sqrt{3}}(1, 0, 1, -1)^T.$$

Part (b): Decomposition of $\mathbf{b}$ Since $\{\mathbf{e}_1, \mathbf{e}_2\}$ is only an orthonormal basis for the **subspace** $W \subset \mathbb{R}^4$ (not all of $\mathbb{R}^4$), the formula $(\mathbf{e}_1 \cdot \mathbf{b})\mathbf{e}_1 + (\mathbf{e}_2 \cdot \mathbf{b})\mathbf{e}_2$ gives the **orthogonal projection** of $\mathbf{b}$ onto W , not $\mathbf{b}$ itself (since $\mathbf{b} \notin W$).

Compute the coefficients:

$$\mathbf{e}_1 \cdot \mathbf{b} = \frac{1}{\sqrt{3}}, \quad \mathbf{e}_2 \cdot \mathbf{b} = \frac{1}{\sqrt{3}}.$$

The projection (component of $\mathbf{b}$ in W):

$$\mathbf{p} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}(1, 1, 0, 1)^T + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}(1, 0, 1, -1)^T = \frac{1}{3}(1, 1, 0, 1)^T + \frac{1}{3}(1, 0, 1, -1)^T = \frac{1}{3}(2, 1, 1, 0)^T.$$

The residual (component of $\mathbf{b}$ orthogonal to W):

$$\mathbf{r} = \mathbf{b} - \mathbf{p} = (1, 0, 0, 0)^T - \frac{1}{3}(2, 1, 1, 0)^T = \frac{1}{3}(1, -1, -1, 0)^T.$$

$$\text{Check: } \mathbf{r} \cdot \mathbf{u}_1 = \frac{1}{3}(1 - 1 + 0 + 0) = 0, \quad \mathbf{r} \cdot \mathbf{u}_2 = \frac{1}{3}(2 - 1 - 1 + 0) = 0.$$

$$\text{Decomposition: } \mathbf{b} = \underbrace{\frac{1}{3}(2, 1, 1, 0)^T}_{\in W} + \underbrace{\frac{1}{3}(1, -1, -1, 0)^T}_{\perp W}.$$

Part (c): Closest vector and distance The closest vector in W to $\mathbf{b}$ is the orthogonal projection:

$$\mathbf{p} = \frac{1}{3}(2, 1, 1, 0)^T.$$

The distance from $\mathbf{b}$ to W is:

$$\|\mathbf{r}\| = \frac{1}{3}\|(1, -1, -1, 0)^T\| = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}.$$

Key concept: Among all vectors $\mathbf{w} \in W$, the projection $\mathbf{p}$ uniquely minimises $\|\mathbf{b} - \mathbf{w}\|$. This follows from Pythagoras: for any $\mathbf{w} \in W$, write $\mathbf{b} - \mathbf{w} = \mathbf{r} + (\mathbf{p} - \mathbf{w})$ where the two parts are orthogonal, so $\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{r}\|^2 + \|\mathbf{p} - \mathbf{w}\|^2 \geq \|\mathbf{r}\|^2$.

Problem 4 — Eigenvalues, Diagonalization, and the Spectral Theorem

Let

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

- (a) Compute the characteristic polynomial $c_A(t)$ and find all eigenvalues.
- (b) For each eigenvalue find a basis for the eigenspace $E_\lambda(A)$.
- (c) Without computing anything further, explain why eigenvectors from different eigenspaces are orthogonal.
- (d) Find an orthogonal matrix P and diagonal matrix D such that $P^{-1}AP = D$.

Solution

Part (a): Characteristic polynomial

$$c_A(t) = \det(tI - A) = \det \begin{pmatrix} t-3 & -1 & -1 \\ -1 & t-3 & -1 \\ -1 & -1 & t-3 \end{pmatrix}.$$

Let $s = t - 3$. Add all rows to row 1 to get the factor $(s - 2)$:

$$= (s - 2) \det \begin{pmatrix} 1 & 1 & 1 \\ -1 & s & -1 \\ -1 & -1 & s \end{pmatrix}.$$

$R_2 \rightarrow R_2 + R_1, R_3 \rightarrow R_3 + R_1$, then expand along the upper triangular form:

$$= (s - 2)(s + 1)^2.$$

Substituting back $s = t - 3$:

$$\boxed{c_A(t) = (t - 5)(t - 2)^2.}$$

Eigenvalues: $\lambda_1 = 5$ (multiplicity 1), $\lambda_2 = 2$ (multiplicity 2).

Part (b): Eigenspaces $\lambda = 5$: Row reduce $A - 5I$:

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

One free variable: basis $\{(1, 1, 1)^T\}$, so $\dim E_5(A) = 1$.

$\lambda = 2$: Row reduce $A - 2I$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Two free variables: basis $\{(-1, 1, 0)^T, (-1, 0, 1)^T\}$, so $\dim E_2(A) = 2$.

Since $\dim E_5 + \dim E_2 = 1 + 2 = 3 = n$, A is diagonalizable.

Part (c): Orthogonality of eigenvectors from different eigenspaces

By the **Spectral Theorem for real symmetric matrices**: if $A = A^T$ and $A\mathbf{u} = \lambda\mathbf{u}$, $A\mathbf{v} = \mu\mathbf{v}$ with $\lambda \neq \mu$, then $\mathbf{u} \perp \mathbf{v}$.

Proof: $\lambda(\mathbf{u} \cdot \mathbf{v}) = (A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^T \mathbf{v}) = \mathbf{u} \cdot (A\mathbf{v}) = \mu(\mathbf{u} \cdot \mathbf{v})$, so $(\lambda - \mu)(\mathbf{u} \cdot \mathbf{v}) = 0$, and since $\lambda \neq \mu$ we get $\mathbf{u} \cdot \mathbf{v} = 0$.

Part (d): Orthogonal diagonalization The 5-eigenvector normalises to $\mathbf{p}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)^T$.

Apply Gram–Schmidt to the basis of E_2 :

Let $\mathbf{w}_1 = (-1, 1, 0)^T$, $\|\mathbf{w}_1\|^2 = 2$.

$$\mathbf{w}_2 = (-1, 0, 1)^T - \frac{1}{2}(-1, 1, 0)^T = \left(-\frac{1}{2}, -\frac{1}{2}, 1\right)^T.$$

Scaling: $(-1, -1, 2)^T$ with norm $\sqrt{6}$.

Normalising:

$$\mathbf{p}_2 = \frac{1}{\sqrt{2}}(-1, 1, 0)^T, \quad \mathbf{p}_3 = \frac{1}{\sqrt{6}}(-1, -1, 2)^T.$$

$$P = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

where P is orthogonal and $P^{-1}AP = D$.

Problem 5 — Projection Matrices (Multi-part Proof)

Let A be an $n \times n$ real matrix satisfying $A^2 = A$ (called a **projection matrix**).

(a) Show that the only possible eigenvalues of A are 0 and 1.

(b) Show that $\mathbb{R}^n = \ker A \oplus \ker(A - I_n)$, i.e. every $\mathbf{v} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \ker A$ and $\mathbf{w} \in \ker(A - I_n)$.

(c) Deduce that A is diagonalizable over $\mathbb{R}$, and state what $\dim(\ker A) + \dim(\ker(A - I_n))$ must equal.

(d) Suppose further that A is symmetric. Prove that $\ker A \perp \ker(A - I_n)$.

Solution

Part (a): Only eigenvalues are 0 and 1 Suppose $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then using $A^2 = A$:

$$\lambda^2\mathbf{v} = A^2\mathbf{v} = A\mathbf{v} = \lambda\mathbf{v}.$$

So $(\lambda^2 - \lambda)\mathbf{v} = \mathbf{0}$. Since $\mathbf{v} \neq \mathbf{0}$, we get $\lambda(\lambda - 1) = 0$, giving $\lambda = 0$ or $\lambda = 1$. $\square$

Important: This only shows 0 and 1 are the only *possible* eigenvalues — it does not guarantee either actually occurs for a given A . We cannot assume $\mathbb{R}^n$ is spanned by eigenvectors without proving it, which is what part (b) does directly.

Part (b): Direct sum decomposition **Existence:** For any $\mathbf{v} \in \mathbb{R}^n$, write:

$$\mathbf{v} = \underbrace{(\mathbf{v} - A\mathbf{v})}_{\mathbf{u}} + \underbrace{A\mathbf{v}}_{\mathbf{w}}.$$

- $\mathbf{u} \in \ker A$: $A\mathbf{u} = A\mathbf{v} - A^2\mathbf{v} = A\mathbf{v} - A\mathbf{v} = \mathbf{0}$
- $\mathbf{w} \in \ker(A - I)$: $(A - I)\mathbf{w} = A^2\mathbf{v} - A\mathbf{v} = A\mathbf{v} - A\mathbf{v} = \mathbf{0}$

Uniqueness: Suppose $\mathbf{v} = \mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$. Then $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$. Call this $\mathbf{z}$. Then $A\mathbf{z} = \mathbf{0}$ (left side in $\ker A$) and $A\mathbf{z} = \mathbf{z}$ (right side in $\ker(A - I)$), so $\mathbf{z} = \mathbf{0}$, giving $\mathbf{u} = \mathbf{u}'$ and $\mathbf{w} = \mathbf{w}'$. $\square$

Part (c): Diagonalizability From part (b), every $\mathbf{v} \in \mathbb{R}^n$ decomposes as a sum of a vector in $\ker A$ and a vector in $\ker(A - I)$. Taking a basis for each subspace and combining gives a basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of A (with eigenvalue 0 or 1 respectively). Hence A is diagonalizable over $\mathbb{R}$.

The two bases together span $\mathbb{R}^n$ and are linearly independent (by uniqueness of the decomposition in part (b)), so:

$$\dim(\ker A) + \dim(\ker(A - I_n)) = n. \quad \square$$

Part (d): $\ker A \perp \ker(A - I)$ when A is symmetric Let $\mathbf{u} \in \ker A$ and $\mathbf{w} \in \ker(A - I)$, so $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{w}$.

$$\mathbf{u} \cdot \mathbf{w} = \mathbf{u} \cdot (A\mathbf{w}) = (A^T\mathbf{u}) \cdot \mathbf{w} = (A\mathbf{u}) \cdot \mathbf{w} \quad (\text{since } A^T = A) = \mathbf{0} \cdot \mathbf{w} = 0.$$

Hence $\mathbf{u} \perp \mathbf{w}$. $\square$

This is the spectral theorem argument applied to eigenvalues 0 and 1, which are distinct — eigenvectors of a real symmetric matrix with different eigenvalues are always orthogonal.